

# MongoDB Data Types Practice

## Prepare the data

```
use retail_db
```

```
db.datatype_practice.insertMany([
  {
    product_id: 501,
    product_name: "Laptop",
    price: 65000,
    stock: 10,
    available: true,
    launch_date: ISODate("2026-01-10"),
    colors: ["Black", "Silver"],
    details: {
      brand: "Dell",
      warranty_years: 2
    }
  },
  {
    product_id: 502,
    product_name: "Headphones",
    price: 4500,
    stock: 25,
    available: true,
    launch_date: ISODate("2026-03-15"),
    colors: ["Black", "Blue"],
    details: {
      brand: "Sony",
      warranty_years: 1
    }
  },
  {
    product_id: 503,
    product_name: "Office Chair",
    price: 9000,
    stock: 0,
    available: false,
    launch_date: ISODate("2025-12-20"),
```

```

        colors: ["Black", "Grey"],
        details: {
            brand: "GreenSoul",
            warranty_years: 3
        }
    },
    {
        product_id: 504,
        product_name: "Smart Watch",
        price: 18000,
        stock: 8,
        available: true,
        launch_date: ISODate("2026-06-01"),
        colors: ["Black", "Red"],
        details: {
            brand: "Samsung",
            warranty_years: 1
        }
    }
])

```

Check:

```
db.datatype_practice.find()
```

## Exercises

1. Display all products.
2. Find the product named `Laptop` .
3. Find products whose price is greater than `10000` .
4. Find products having stock less than `10` .
5. Find products where:

```
available = true
```

6. Find products where:

`available = false`

7. Find products launched after:

`1 January 2026`

8. Find products available in the color `Black` .

9. Find products available in both:

`Black`

`Blue`

10. Find products where the brand is `Samsung` .

11. Find products having more than 1 year warranty.

12. Increase the stock of product `502` by `5` .

13. Change product `503` from unavailable to available.

14. Add the color `White` to product `501` .

15. Change the warranty of product `504` to `2` years.

16. Display only:

`product_name`

`price`

17. Find the datatype of the `price` field using `$type` .

18. Find the datatype of the `available` field.

19. Find the datatype of the `launch_date` field.

20. Find the datatype of the `colors` field.